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(54) **WEIGHT-DISTRIBUTED CUTLERY AND
RELATED METHODS**

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(71) Applicant: **Bruce Campbell**, Ashland, OR (US)

(72) Inventor: **Bruce Campbell**, Ashland, OR (US)

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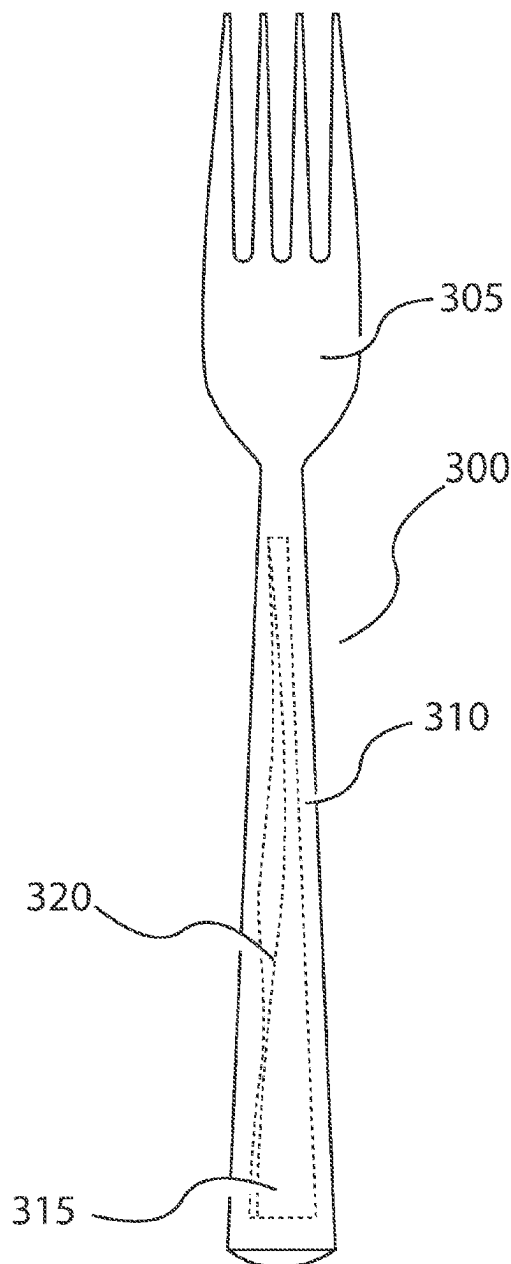
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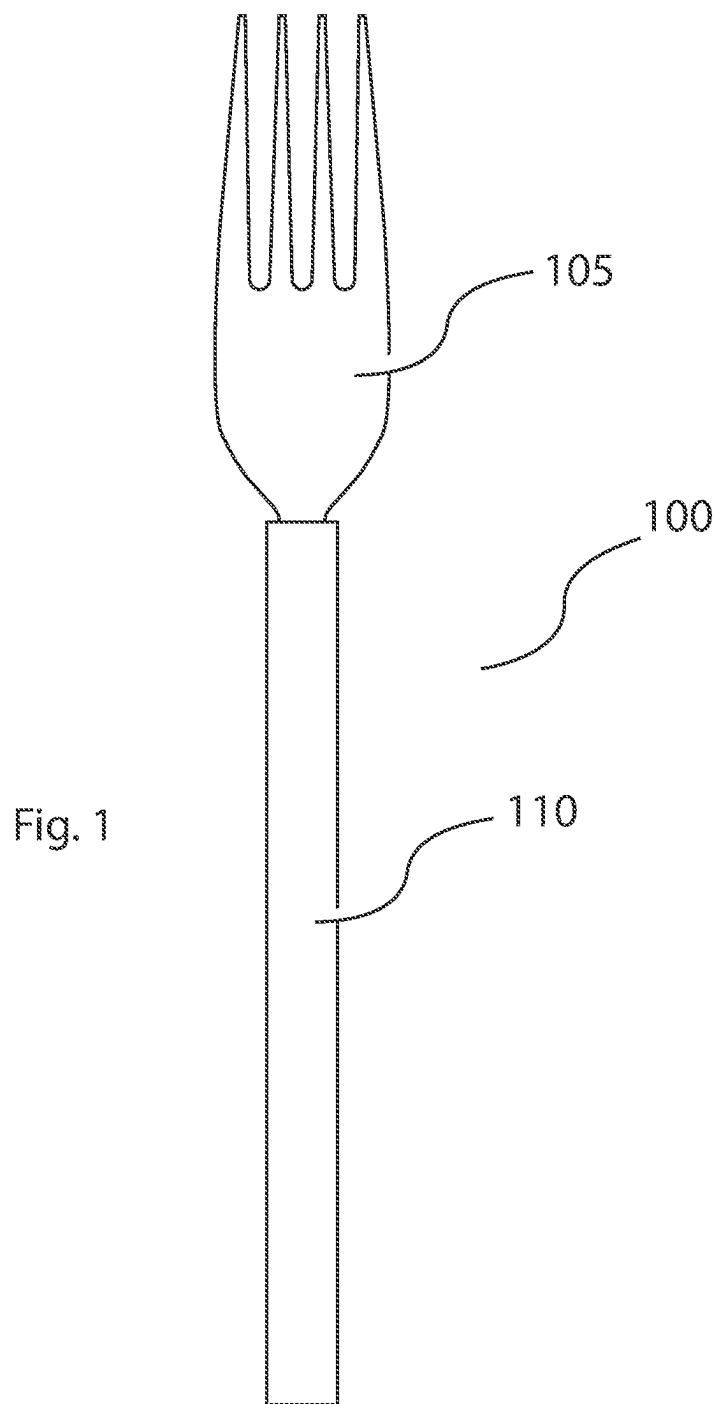
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ABSTRACT

The utensil includes a functional section and a handle, with the functional section being heavier than the handle. The handle is made of a lighter weight material than the functional section and may have a hollow center or a minimal design to reduce weight. The handle may also integrate a small amount of liquid to balance the center of gravity. The handle and functional section can be made of different materials or the same type of material. A cuff section may bridge the gap between the handle and functional section.





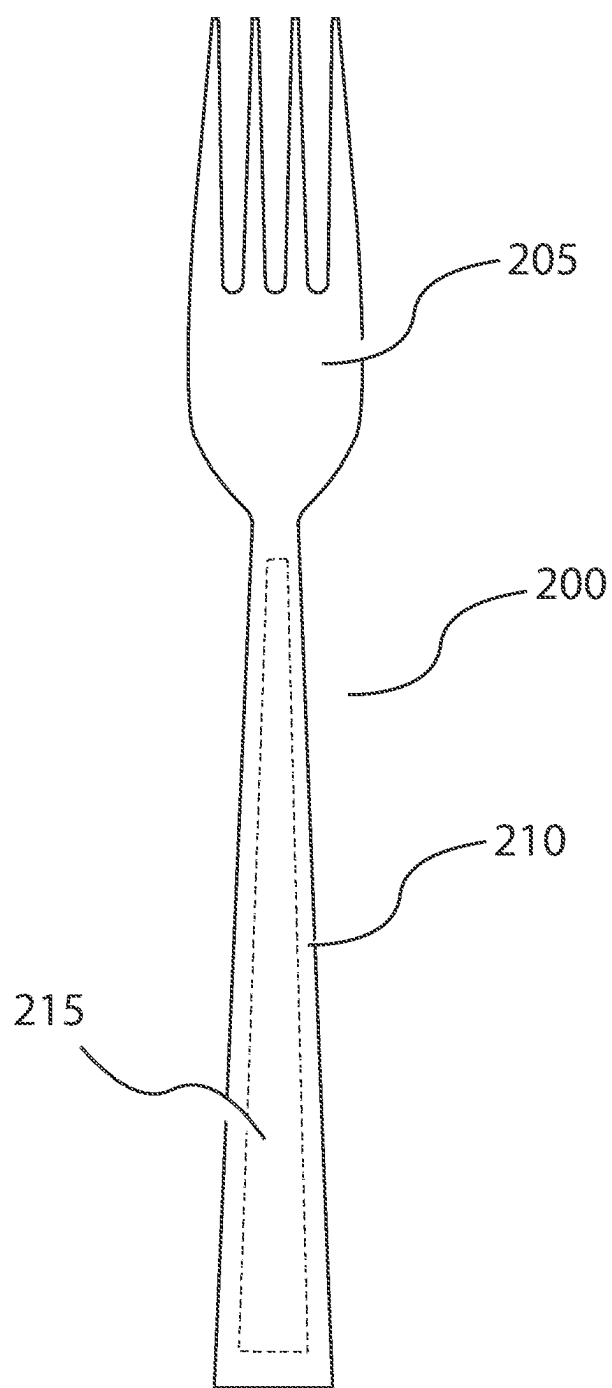


Fig. 2

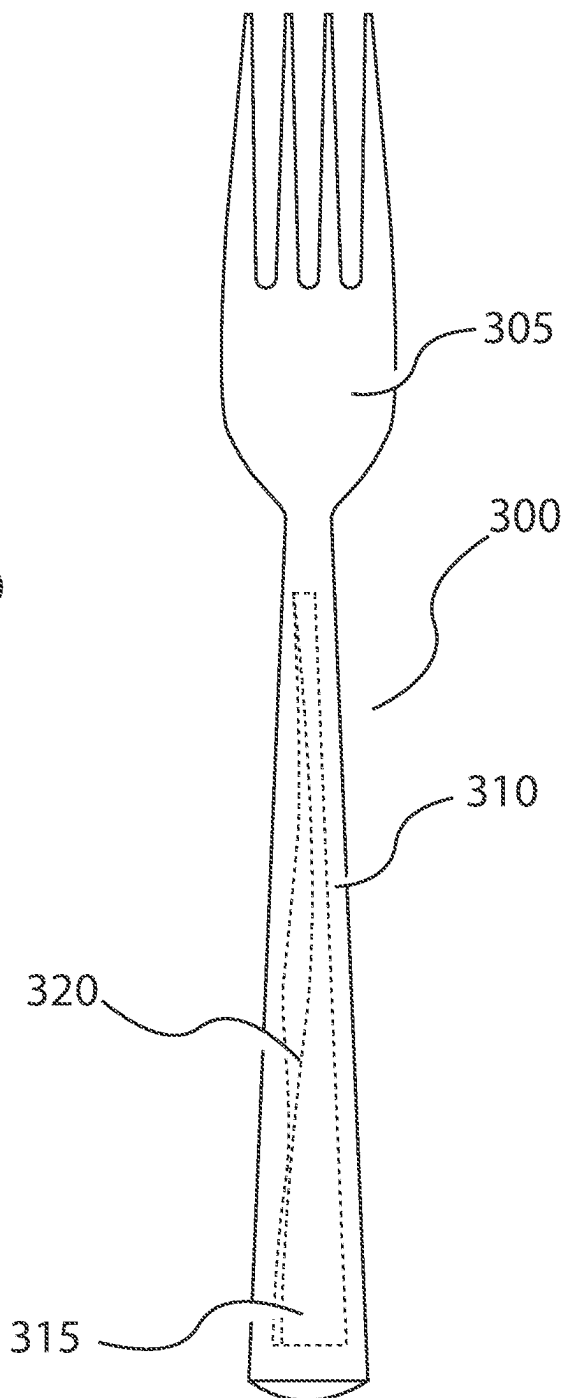


Fig. 3

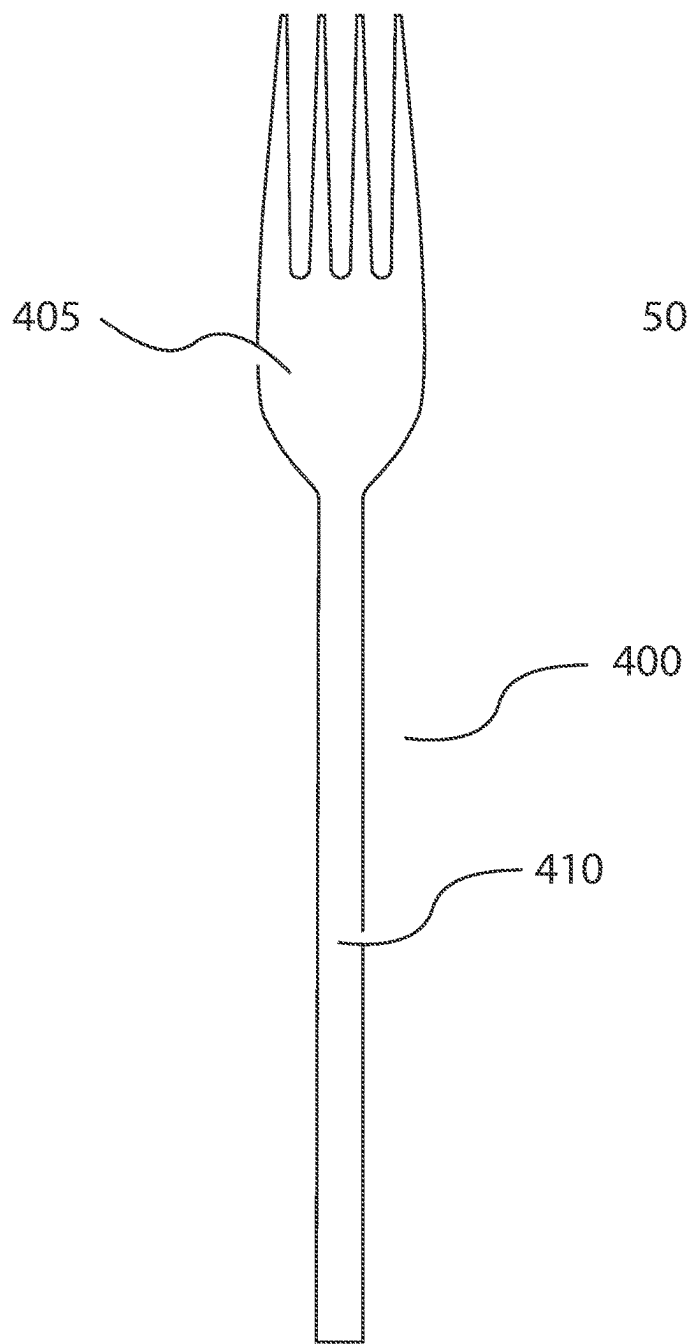


Fig. 4

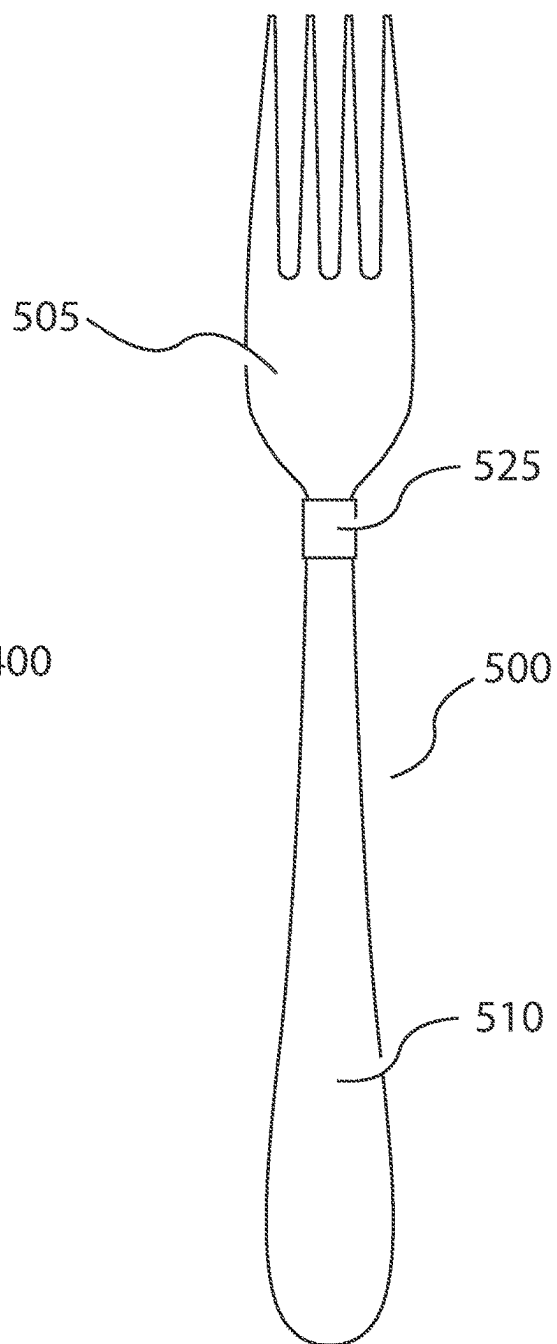


Fig. 5

WEIGHT-DISTRIBUTED CUTLERY AND RELATED METHODS

GOVERNMENT CONTRACT

[0001] Not applicable.

CROSS-REFERENCE TO RELATED APPLICATIONS

[0002] Not applicable.

STATEMENT RE. FEDERALLY SPONSORED RESEARCH/DEVELOPMENT

[0003] Not applicable.

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TECHNICAL FIELD

[0005] The invention pertains to the design of cutlery, specifically focusing on the weight distribution between the functional section and the handle of the utensil.

BACKGROUND

[0006] The field of cutlery design has seen various innovations over the years, aimed at improving the user experience. One of the persistent challenges in this field is the issues relating to use of utensils in an environment where such cutlery interacts with other objects, such as plates and tables, and when slightly misplaced, a utensil may fall off of a plate or table causing a mess or embarrassment by the user of the utensil.

[0007] One problem that may cause such issues is an imbalance in the cutlery that can lead to inconvenience during use, such as the utensil falling off plates or tables. The improper balancing of the weight distribution may cause utensils to tip over when a functional section is placed on a plate and a utensil handle is hanging off such plate or object. This issue is prevalent in various settings, from home kitchens to restaurants, and can lead to spills and messes. Therefore, there is a need for improved designs in cutlery that address this weight distribution issue.

SUMMARY

[0008] Exemplary embodiments of the present invention relate to weight distributed utensil apparatus, and more particularly to weight distributed utensils that are capable of balancing the weight of the functional section of the utensil on a plate or other objects such as, for example tabletops.

[0009] In accordance with certain embodiments, a utensil is provided that may be comprising a functional section and a handle, wherein the functional section may be configured to be heavier than the handle section of the apparatus. The handle may be made of a lighter weight material than the

functional section, which may also have a hollow center or a minimal design to reduce weight and distribute weight in a manner that prevents utensils of all types from falling off of other objects when resting.

[0010] In some embodiments, the handle may also integrate a small amount of liquid to balance the center of gravity when being used during meals or throughout the cooking process. In many embodiments, the utensil may comprise a functional section and a handle, wherein the functional section is heavier than the handle. The handle and functional section may be made of the same type of material, with the handle being hollow to reduce weight. The handle may also integrate a small amount of liquid to balance the center of gravity.

BRIEF DESCRIPTION OF THE FIGURES

[0011] FIG. 1 shows a front view of an embodiment of the present invention having a multi-material design;

[0012] FIG. 2 shows a front view of an embodiment of the present invention having a hollow interior;

[0013] FIG. 3 shows a front view of an embodiment of the present invention having a hollow interior and an amount of liquid present in such hollow interior;

[0014] FIG. 4 shows a front view of an embodiment of the present invention configured with a handle having a narrow circumference; and

[0015] FIG. 5 shows a front view of an embodiment of the present invention that integrates a cuff feature.

DETAILED DESCRIPTION

[0016] The present invention is relating to cutlery that has a weight distribution wherein the blade, spoon bowl, or fork and prongs portion of a utensil (collectively referred to herein as the “functional section”) that is configured to be heavier than the handle section of the cutlery or utensil.

[0017] In some embodiments, such as the one depicted in FIG. 1, there may be a mixed material design wherein the handle 110 of the utensil 100 may be made of a material that is of a lighter weight than the functional section 105 of the utensil. Such embodiments may utilize, for example, a lightweight wood or similar material for the handle 110, and the functional section 105 may be made of, for example, stainless steel or another metallic material. Additionally, the handle 110 may be a lightweight material that also has a hollow center to further reduce the weight of the handle 110.

[0018] In additional embodiments, the utensil 100 may be made entirely of the same type of material. In many embodiments, to reduce the weight for proper distribution (handle being heavier than the functional section), for example, as shown in FIG. 2, the handle 210 of the utensil 200 may be, for example, a hollow metal handle having a hollow section 215, wherein the reduction of solid material inside of the handle 210 may balance the utensil 200 in a way that the handle 210 is lighter than a traditional utensil handle as well as being lighter than the functional section 205 that may be the same material as the handle but may be configured to be solid all the way through the functional section 205.

[0019] In similar embodiments, the hollow handle of the utensil may also integrate a small amount of liquid. By introducing a small amount of liquid, the design may not increase the weight of the handle significantly enough to

make it heavier than the functional section and it may also balance the center of gravity for a better experience when the utensil is tilted or in use.

[0020] In many embodiments of the present invention, the functional section **105** and the handle **110** may each be made of differing types of metallic material and in many of such embodiments, the metallic material of the functional section **105** may be of a greater density than the metallic material of the handle **110**.

[0021] In some embodiments, the utensil may integrate a design that has a minimal handle design wherein the circumference of the handle is less than the traditional handle design, so the weight is reduced in comparison to the functional section.

[0022] In certain embodiments the present invention may integrate multiple types of metallic or other solid materials such as embodiments depicted by FIG. 5, where, for example, stainless steel may be used for the functional section **505** and a lighter weight material such as, for example, titanium may be used for the handle **510** so the weight of the handle may remain lighter on the handle **510** than the functional section **505**. In such embodiments, the metals may differ in look and feel, so the present invention may be configured to utilize a cuff section **525** designed to bridge the gap between the handle **510** and functional section **505**. The cuff section **525** may be used to transition between the handle **510** and functional section **505** and may also be configured to serve as a securing mechanism that secures or bonds the handle **510** and functional section **505** together while also serving to create a more aesthetic look and feel between the differing metal or material types.

[0023] Referring to FIG. 1, in accordance with certain embodiments, the present disclosure may describe a multi-functional utensil **100**. The utensil **100** may comprise components: a functional section **105** and a handle **110**. The functional section **105**, which may manifest as a fork, spoon, knife, or any other utensil head, is fabricated from materials selected for durability and functionality suitable for its intended use. The handle **110**, on the other hand, is constructed from a lightweight material which could be wood, bamboo, or any other lightweight yet sturdy material, providing a distinct advantage in terms of the overall weight distribution and ergonomics of the utensil **100**.

[0024] It should be understood that the material composition of the handle **110** is not limited to wood or bamboo but may include a variety of lightweight materials capable of offering the requisite strength and comfort for the user. This design consideration ensures that the handle **110** remains significantly lighter in weight compared to the functional section **105**. The choice of material for the handle **110** may be guided by factors such as environmental sustainability, cost-effectiveness, and user preference, among others.

[0025] The differential in weight between the handle **110** and the functional section **105** presents several advantages. Firstly, it may contribute to an improved balance of the utensil **100**, whereby the center of gravity is closer to the functional section **105**. This configuration could potentially enhance the user's control and precision during use, whether cutting, scooping, or spearing food items, thereby enhancing the overall user experience.

[0026] Additionally, the lighter weight of the handle **110** may reduce the overall weight of the utensil **100**, making it more comfortable for extended use. This could be particularly beneficial in settings where utensils are used frequently

and for prolonged periods, such as in restaurants or households. The reduced weight may also be advantageous for individuals with limited hand strength or dexterity, including the elderly or those with certain disabilities, facilitating easier handling and use.

[0027] Furthermore, the use of lightweight materials for the handle **110** could offer environmental benefits, especially if sustainable materials like bamboo are utilized. Such materials often have a lower carbon footprint compared to traditional plastics or metals typically used in utensil handles, aligning the design of the utensil **100** with eco-friendly practices and preferences of environmentally conscious consumers.

[0028] It is to be appreciated that the specific design and material selection for the functional section **105** and the handle **110** of the utensil **100** may vary depending on the intended application, user needs, and manufacturing considerations. The embodiments described herein are intended to be illustrative rather than limiting, as numerous variations of the utensil **100** are conceivable without departing from the scope of the invention.

[0029] In many embodiments of the present invention, the utensil **200** may be characterized by its unique construction, comprising a handle **210**, a functional section **205**, and a hollow section **215** within the handle **210**. Both the handle **210** and the functional section **205** may be fabricated from a type of metal, offering durability and a premium feel. However, it is within the field of the invention that the type of metal selected for the handle **210** and the functional section **205** could vary, allowing for customization based on specific requirements such as aesthetic preference and weight considerations.

[0030] In some embodiments, such as those illustrated in FIG. 2, the handle **210** and the functional section **205** may be constructed from the same type of metal. Such uniform material choice may not only enhance the aesthetic appeal of the utensil **200** but also simplify the manufacturing process. Despite the shared material, a distinctive feature of the handle **210** may be a hollow section **215** in the interior of the handle **210**. This hollow section **215** within the handle **210** may be designed to reduce the overall weight of the handle, thereby influencing the weight distribution and ergonomics of the utensil **200**.

[0031] The incorporation of a hollow section **215** in the handle **210** may serve several advantageous purposes. For example, it may contribute to a lighter weight of the handle **210** in comparison to the functional section **205**. This deliberate weight differentiation enhances the balance of the utensil **200**, making it more user-friendly and easier to manipulate during various culinary tasks. Such a design consideration is likely to improve the handling comfort for the user, particularly during prolonged periods of use or for individuals with specific ergonomic needs. Additionally, such weight distribution prevents issues relating to falls and spills when the utensil **200** is placed on the edge of, for example a plate wherein the functional section **205** is placed on the plate or tabletop and the handle **210** is hanging off of such object.

[0032] Moreover, the reduction in material usage for the handle **210**, achieved through the incorporation of the hollow section **215**, may also have implications for cost and environmental sustainability. By utilizing less metal in the handle **210**, the production of the utensil **200** could become more resource-efficient and potentially lower in cost, mak-

ing it accessible to a broader market segment. Additionally, the minimized use of metal aligns with eco-friendly manufacturing practices, reducing the environmental footprint associated with the production of the utensil **200**.

[0033] In many embodiments of the present invention, the functional section **205** and the handle **210** may each be made of differing types of metallic material and in many of such embodiments, the metallic material of the functional section **105** may be of a greater density than the metallic material of the handle **210**.

[0034] It is imperative to understand that the precise dimensions, shape, and design of the hollow section within the handle **210** can vary according to specific embodiments of the invention. These variations may be influenced by factors such as the type of metal used, the intended use of the utensil **200**, and ergonomic considerations. The detailed embodiments described herein are intended for illustrative purposes and should not be construed as limiting the scope of the invention. Variations and modifications to the described embodiments are within the purview of those skilled in the art, without departing from the essence of the invention as defined by the appended claims.

[0035] In some embodiments of the present invention, as depicted in FIG. 3, the fundamental structure of the previously described utensil **200**, comprising a handle **310**, a functional section **305**, and a hollow section **315** remain, but also may include a liquid integration **320** within the hollow section **315**. In many embodiments, such configuration may be constructed predominately from metals or metal alloys, selected for their durability, aesthetic appeal, and suitability for the utensil's intended culinary functions. Additionally, such configuration may be achieved using various other materials, such as, ceramics, glass, advanced plastics, silicon, or other materials that exhibit durability, resistance to corrosion or liquid leakage, and the capacity to retain their structural integrity under the conditions of use typically encountered by utensils.

[0036] A feature of the utensil **300**, may be the inclusion of such internal liquid **320** within the hollow section **315** of the handle **310**. The hollow section **315**, akin to the utensil **200** design, serves to reduce the overall weight of the handle **310**. However, the strategic addition of the internal liquid **320** within this hollow section **315** introduces a novel approach to balancing the utensil **300** during use while also providing the balance necessary to prevent falls and spills of the various embodiments of the present invention.

[0037] The internal liquid **320** may be carefully calibrated in volume and liquid density to achieve a desired balance effect without significantly increasing the weight of the utensil **300**. The presence of such liquid may afford the utensil **300** a dynamic balance, allowing it to adjust to various angles and positions while being manipulated by the user. This can potentially enhance the utensil's ergonomic characteristics, making it more comfortable and intuitive to use across a wide range of dining and cooking activities.

[0038] The concept of incorporating an internal liquid **320** for balance is predicated on the principle of fluid dynamics, where the movement of the liquid within the hollow section **315** can counterbalance shifts in the utensil's orientation. This may particularly benefit precision tasks that require stability and control, such as cutting or intricate food plating.

[0039] It is to be appreciated that the specific composition of the internal liquid **320**, as well as the volume and the design of the hollow section **315**, may vary according to

different implementations of the utensil **300**. Such variations could be influenced by the type of food for which the utensil is intended, the specific user demographic targeted, and manufacturing considerations.

[0040] In many embodiments of the present invention, the functional section **405** and the handle **410** may each be made of differing types of metallic material and in many of such embodiments, the metallic material of the functional section **405** may be of a greater density than the metallic material of the handle **410**.

[0041] Such embodiments illustrate an inventive scope of the disclosed utensil design, demonstrating the flexibility and adaptability of the concept to incorporate features that enhance user experience. While the embodiments described herein focuses on the addition of an internal liquid **320** for balance, it is understood that this feature is merely illustrative and may be made up of other materials that provide the balance, such as, for example, sand, metallic shavings or fragments, among other materials that may move within an open space due to gravity. Other modifications and variations of the utensil **300**, including but not limited to the material composition, shape, and size of the functional section **305** and the handle **310**, are envisioned within the scope of the invention, as defined by the appended claims.

[0042] In many embodiments of the invention, such as those depicted in FIG. 4, the design considerations of the previously outlined utensil configurations, including a handle **410** and a functional section **405** may be contemplated. For example, both the handle **410** and the functional section **405** may be fabricated from materials selected for their robustness, functionality, and compliance with food safety standards.

[0043] Additionally, such embodiments of the utensil **400** may be configured with a reduced handle **410** circumference. This design decision aims to decrease the overall weight of the handle **410**, ensuring that it is lighter than the functional section **405**. The reduction in circumference may be calculated to not compromise the structural integrity of the utensil **400** during its intended use. This balance between reduced weight and maintained strength addresses a concern among users regarding the durability and ergonomic viability of lightweight utensils. Such smaller circumference may have a uniform measurement, that may extend from the portion of the handle **410** where it is connected with the functional section **405**. For example, the diameter of the handle **410** may the same width throughout the entire length of the handle **410** from a distal end to the connected end.

[0044] The smaller circumference of the handle **410** may offer several advantages. Primarily, it may lead to a utensil that is easier to grip and maneuver, particularly for users with smaller hands or those seeking a more delicate tool for precise culinary tasks. Despite its reduced size, the handle **410** is engineered to retain a strength that resists bending, breaking, or deforming under normal usage conditions. This is achieved through the selection of high-strength materials, advanced manufacturing techniques, or a combination of both, which allows for a slender yet sturdy handle design.

[0045] Moreover, the strategic reduction in handle circumference could have environmental and economic benefits. Less material usage in the handle **410** may result in a decrease in the raw materials required for the utensil's production, potentially lowering manufacturing costs and reducing the environmental footprint associated with material extraction and processing.

[0046] It should be understood that the dimensions and material composition of the handle **410** and the functional section **405** are variable, depending on the specific application, user preferences, and manufacturing considerations. This embodiment, therefore, serves as an illustrative example of the utensil **400**'s design flexibility, highlighting the potential for customization and adaptation to meet diverse needs.

[0047] The described embodiment exemplifies the inventive approach taken in developing the utensil **400**, showcasing the feasibility of combining reduced weight with uncompromised strength and structural integrity. Such innovations underscore the commitment to enhancing the user experience through ergonomic design, material efficiency, and sustainability. While this description focuses on a particular aspect of handle design, it is to be appreciated that numerous other modifications and variations of the utensil **400** are conceivable within the scope of the invention, as outlined in the appended claims.

[0048] In many embodiments, as shown in FIG. 5, the design of the utensil **500** may encompass a functional section **505** and a handle **510**, each may be fabricated from distinct metals chosen for their weight and durability properties. The functional section **505** may utilize, for example a heavier metal, such as stainless steel, known for its durability, corrosion resistance, and ease of maintenance. Conversely, the handle **510** may be constructed, for example, from a lighter metal, such as titanium, which is known for its strength-to-weight ratio, corrosion resistance, and hypoallergenic properties.

[0049] An additional feature of some embodiments of the present invention is the incorporation of a cuff **525**, situated at the juncture between the handle **510** and the functional section **505**. The cuff **525** may serve dual purposes. Such as the ability to function as an aesthetic element designed to seamlessly integrate the visual transition between the two differing metals, providing a cohesive and polished appearance to the utensil **500**. This consideration may ensure that the utensil **500** maintains an elegant and unified design, despite the use of distinct materials.

[0050] Furthermore, the cuff **525** may play a crucial structural role, acting as a bonding agent between the handle **510** and the functional section **505**. This could be achieved through various means, such as welding, adhesion, mechanical fastening, or a combination thereof, depending on the specific materials used and the desired outcome in terms of durability and manufacturing efficiency. The choice of bonding method may also consider factors such as ease of assembly, cost-effectiveness, and the impact on the overall aesthetic of the utensil **500**.

[0051] The use of a material such as, for example, stainless steel for the functional section **505** may ensure that the utensil **500** is capable of withstanding the rigors of everyday use, including exposure to high temperatures, acidic foods, and frequent washing, without compromising its functional integrity or aesthetic appeal. In certain embodiments where, for example, titanium is selected for the handle **510**, may contribute to a significant reduction in the utensil **500** overall weight as well as a reduced handle **510** weight, enhancing user comfort and reducing fatigue during extended periods of use.

[0052] In many embodiments of the present invention, the functional section **505** and the handle **510** may each be made of differing types of metallic material and in many of such

embodiments, the metallic material of the functional section **505** may be of a greater density than the metallic material of the handle **510**.

[0053] It should be appreciated that the selection of stainless steel and titanium for the functional section **505** and the handle **510**, respectively, exemplifies just one possible material combination within the scope of the invention. The integration of a cuff **525** to both aesthetically and structurally unite these components further underscores the innovative aspect of the utensil **500**'s design.

[0054] Such embodiments, as illustrated in FIG. 5, provide examples of how differing materials can be utilized in utensil design to achieve specific objectives, such as enhanced durability, reduced weight, and improved ergonomic and aesthetic qualities. While the detailed description focuses on the use of stainless steel, titanium, or metallic materials coupled with a cuff **525**, it is to be understood that various modifications and variations of the utensil **500** are conceivable, remaining within the essence and scope of the invention as defined by the appended claims.

What is claimed is:

1. A balanced utensil apparatus, comprising: a functional section and a handle, wherein the functional section is configured to be heavier than the handle; wherein the handle is made of a lighter weight material than the functional section; wherein the handle is configured to contain a hollow section within the handle; wherein the hollow section contains a volume of liquid less than the volume of the hollow section; and wherein the configuration of the apparatus results in the ability to place the functional section on an object with the handle extending off the object without falling.
2. The apparatus of claim 1, wherein the handle is made of a lightweight wood or a similar material.
3. The apparatus of claim 1, wherein the functional section and the handle are each made of differing types of metallic material and wherein the metallic material of the functional section is of a greater density than the metallic material of the handle.
4. The apparatus of claim 1, wherein the handle has a hollow center.
5. The apparatus of claim 1, wherein the handle and functional section are made of the same materials.
6. The apparatus of claim 4, wherein the hollow section of the handle contains a volume of liquid less than the volume of the hollow section.
7. The apparatus of claim 6 wherein the hollow section of the handle is exposed on a distal end of the handle.
8. The apparatus of claim 1, wherein a cuff is configured to wrap around a portion of both the handle and the functional section.
9. The apparatus of claim 8, wherein the cuff section is used to secure the handle and functional section together.
10. A balanced utensil apparatus, comprising: a functional section and a handle, wherein the functional section is configured to be heavier than the handle; wherein the handle is made of the same material as the functional section; wherein the handle is configured to contain a hollow section within the handle; wherein the configuration of the apparatus results in the ability to place the functional section on an object with the handle extending off the object without falling.

11. The apparatus of claim **10**, wherein the hollow section of the handle contains a volume of liquid less than the volume of the hollow section.

12. The utensil of claim **10**, wherein the hollow section of the handle is exposed on a distal end of the handle.

13. The utensil of claim **10**, wherein a cuff is configured to wrap around a portion of both the handle and the functional section.

14. The utensil of claim **10**, wherein a cuff section is used to secure the handle and functional section together.

15. A balanced utensil apparatus, comprising:
a functional section and a handle, wherein the functional section is configured to be heavier than the handle;
a cuff configured to adjacent to both the functional section and the handle;

wherein the handle is made of a lighter weight material than the functional section;

wherein the cuff is configured to bond the functional section and the handle together;

wherein the configuration of the apparatus results in the ability to place the functional section on an object with the handle extending off the object without falling.

16. The apparatus of claim **15**, wherein there is a hollow section within the handle.

17. The apparatus of claim **16**, wherein the hollow section contains a volume of liquid less than the volume of the hollow section.

18. The utensil of claim **17**, wherein the hollow section of the handle is exposed on a distal end of the handle.

19. A balanced utensil apparatus, comprising:

a functional section and a handle, wherein the functional section is configured to be heavier than the handle;
wherein the handle is configured with a uniform circumference from a connected portion of the functional section to the distal end of the handle; and

wherein the configuration of the apparatus results in the ability to place the functional section on an object with the handle extending off the object without falling.

20. The apparatus of claim **19**, wherein the functional section and the handle are each made of differing types of metallic material and wherein the metallic material of the functional section is of a greater density than the metallic material of the handle.

* * * * *